SAMPLE

Famous Astronomers: Kip Thorne

Gravitational Waves and Black Holes

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Introduction

Some scientists are known for a single equation, others for a single instrument, and a very few for having bet their whole working life on an idea the rest of the field considered unreachable, and then living to see the idea arrive. The astronomer this volume is about belongs emphatically to the last group: a theoretical physicist from a Utah farm town who spent half a century calculating what black holes and gravitational waves ought to look like, who co-founded the enormous instrument that finally caught a wave, and who stood on a stage in 2016 when his field's longest-running prediction came true as a little rising whistle in a detector's data. He is the series' youngest subject so far, close enough to our own day that many of his colleagues are still working, which makes this book both easier and harder to write: the record is rich, and the judgment is still settling.

The approach is the catalog's standing one: the person treated as a working astronomer, not a monument. The chapters follow the mechanics of the career, the science the subject actually did, the instruments and arguments that carried the work, and the honest accounting of what remains open. For this subject the science is the two most extreme objects the universe offers, black holes and gravitational ripples in space itself, and the work is the fifty-year campaign to move both from speculation to laboratory fact. The campaign succeeded within the subject's own lifetime, which almost never happens, and the way it succeeded is the best story the series has yet had to tell.

The one-paragraph preview for the impatient: a boyhood eclipse turned a Utah child toward the sky; a Princeton doctorate under the field's great teacher put him inside the golden age of black hole theory; a career at Caltech made him the field's map-maker, predicting what the sky's violent collisions should look like through a gravitational lens; the mapping turned into an instrument, the instrument took decades and a billion dollars and outlasted its skeptics, and in 2015 it recorded two black holes collosing more than a billion years ago; the Nobel followed; and the subject, well past retirement age, went on to put real black hole physics into a Hollywood movie. The chapters ahead unpack each clause of that sentence.

Chapter 1: The Discovery of Kip Thorne

Every astronomer's story has a first moment, and the subject of this volume told his own plainly: he was eight years old, in the Idaho? no, in the mountain west of his childhood, and an eclipse of the sun darkened the afternoon. He asked his mother to explain it, and her explanation satisfied him for a while, and when it stopped satisfying him he began to work the problem out for himself, and by eleven he had organized the neighborhood children into a course of astronomy lectures complete with reading assignments. A childhood like that, the reader will recognize, rarely turns anywhere else.

The setting deserves its paragraph. He was born in 1940 in a small university town in the northern Utah mountains, the second of five children, into an academic family: his father was a professor of soil science who studied how plants take up minerals from the ground, and his mother was an economist who had put her own doctorate on hold to raise the family and who spent decades championing women's right to return to education, becoming in her sixties a fixture of the university and a co-author of a book on the economics of the household. The family were members of the region's predominant church, and the boy was raised in it; as a young man he quietly left the faith, one of the few personal lines he has spoken of publicly, and with characteristic precision: he said he found himself unable to believe, not that he had been wronged. The household's culture was argument and evidence, and the house rule the children remember is the one the career later ran on: any claim is welcome, if it survives being questioned.

The schooling was fast. He reached the California Institute of Technology as an undergraduate at seventeen, took his bachelor's degree there in 1962, and might have stayed, but the strongest pull in theoretical physics at that moment sat in Princeton, where an older scholar was remaking the teaching of gravity, and the young man moved east to write his doctorate under him. The supervisor's name belongs to the series' volume on the golden age of relativity, and the student's own summary is enough here: the man taught his students to treat the universe as a set of solvable puzzles, and to write about them in language a bright non-specialist could follow. The doctorate, finished in 1965 when the student was twenty-five, was a piece of mathematical machinery for describing waves of gravity of a particularly simple kind, and it set the pattern of everything after: find the clean formulation, make it usable, hand it to others.

Two encounters in those student years belong in the discovery chapter because they shaped the career's texture. The first was the Princeton school itself: the golden age of black hole research was at its height, and the graduate

student was in the room while the field's founding arguments were settled. The second was a year earlier and three thousand miles west: as a Caltech undergraduate he had sat in on a legendary course on gravity taught by the campus's most famous lecturer, a showman-physicist whose notes from those lectures the student would later help see into print decades afterward. The lecturer's own approach to gravitational waves, dismissive and funny and in the end correct, stayed with the student for life, and one of this book's later chapters tells the detective story the student eventually undertook into that lecturer's most famous argument. He returned to Caltech as a researcher in 1967 and never left the institute again except as a visitor; by 1970 he was a professor, and the campus, the field, and the man entered a fifty-year partnership.

The discovery chapter's closing point is the one the series likes to make about every subject: the astronomer was discovered by the sky young, in an ordinary afternoon's darkening, and the difference between him and the reader is not the moment but the fifty years of appointments kept afterward.

A paragraph on the mentor at Princeton, the figure behind the doctorate. The graduate student arrived at the eastern gravity school at the moment it was remaking the field, and his advisor there was the field's great systematizer, the physicist who had named the black hole and rallied a generation around Einstein's hardest legacy; the apprenticeship was demanding and broad, the student drilled in the discipline of asking large questions carefully, and the advisor's habit of issuing bold problems to young people became the younger man's inheritance, the style of a scientific house passed down. The origin note: the Utah student became a relativist at Princeton, formed by a teacher who believed the subject's frontiers were a place for beginners, and the file should name the lineage honestly, because the subject's greatest modern popularizer was himself the product of a chain of mentorship, the student of the man who gave the black hole its name.

A second origin note, on the two inheritances, the faith and the physics, carried together from the mountain west. He grew up in a devout household, the son of a farming and academic family rooted in the Mormon culture of northern Utah, and he pursued physics without rupture: the fields, in his telling, answered different questions, the one the shape of the moral life, the other the shape of spacetime, and he kept both, a working scientist who remained an active participant in his inherited tradition; the coexistence, unremarkable to him, has been an exhibit for a wider conversation about science and religion. The origin's closing note: the file should present the balance without drama, the biography's honest outline being a devout scientist rather than a story of conversion in either direction, and the wider interest of the pairing belongs to the reader to draw, the subject's own account being simply that he never found the two inheritances at war.